THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 06

Full Syllabus (Chapters 1–8) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 5 units	$\sqrt{(4-(1))^2+(6-(2))^2}=\sqrt{9+16}=5$
2	**(d)** $(2,\ 6)$	reflection in the y-axis changes the sign of the x-coordinate
3	**(a)** $(3,\ -7)$	$B=(2x_M-x_A,\ 2y_M-y_A)$
4	**(a)** $5n+2$	common difference 5, so $a_n=5n+2$
5	**(b)** 4	substitute $x=-1$ to get 4
6	**(a)** $\frac{1}{6}$	standard method ⇒ $\frac{1}{6}$
7	**(b)** rational	repeating decimal $=\frac{1}{3}$
8	**(d)** division by zero (undefined)	division by 0 is not defined
9	**(a)** $p+2q$	$a=p,\ b=2q$
10	**(d)** $x^2+15x+56$	$(x+a)(x+b)=x^2+(a+b)x+ab$
11	**(b)** 10 cm	half-chord $=\sqrt{13^2-12^2}=5$, so chord $=2\times5=10$
12	**(d)** 57°	opposite angles of a cyclic quadrilateral add to 180°
13	**(b)** infinitely many	centre can be anywhere on the perpendicular bisector
14	**(c)** 84 cm^2	$s=24$, area $=\sqrt{24\cdot14\cdot7\cdot3}=84$
15	**(c)** 22 cm	$2\pi r=2\times\frac{22}{7}\times3.5=22$
16	**(d)** 180 cm^2	area $=bh$
17	**(d)** $\frac{1}{3}$	favourable outcomes: 3, 6 ⇒ 2 out of 6
18	**(a)** 0.23	$46/200=0.23$
19	**(d)** $n=69$	$2n-1=137\Rightarrow n=69$
20	**(c)** 48	$a+(n-1)d=3+9\times(5)=48$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$AB=\sqrt{4+4}=2\sqrt{2},\ BC=2\sqrt{2},\ AC=4$. $AB=BC$, and $AB^2+BC^2=8+8=16=AC^2$: a right-angled isosceles triangle.	1+1
22	Amount $=200+50m$. $200+50m=750\Rightarrow m=11$ matches.	1+1
23	$x=0.\overline{9}$; $10x=9.\overline{9}$; $10x-x=9\Rightarrow x=1$.	1+1
24	$(x-3)\,(x-2)$	2
25	$OA=OB$ and $\angle AOB=60^\circ$ ⇒ $\triangle OAB$ is equilateral, so $AB=12$ cm.	1+1
26	The hexagon's perimeter is $6r=3\times(2r)=3D$, and the circle is longer than the polygon inside it, so $C>3D$, i.e. $\pi>3$.	1+1
27	(i) $\frac{4}{52}=\frac{1}{13}$. (ii) $\frac{2}{52}=\frac{1}{26}$.	1+1
28	$t_n=11+(n-1)(-3)=14-3n$ (1). Recursive: $t_1=11,\ t_n=t_{n-1}-3$ (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	$A'(-3,4),\ D'(-7,1),\ M'(-9,6)$. $AD=\sqrt{16+9}=5$; $A'D'=\sqrt{(-7+3)^2+(1-4)^2}=\sqrt{16+9}=5$. Reflection preserves lengths.	1+1+1
30	First month ₹500; second ₹650… $a_n=500+150(n-1)=350+150n$. $350+150n=1850\Rightarrow n=10$.	1+1+1
31	Removing a debt of ₹5 from 10 rupees of wealth makes you ₹5 richer: $10-(-5)=10+5=15$. In Brahmagupta's language, taking away a debt is like adding a fortune.	1+2
32	LHS $=2a^2+2b^2$ ✓. With $a=100, b=5$: $105^2+95^2=2(10000+25)=20050$.	1+1+1
33	$\pi(14^2-7^2)=\frac{22}{7}\times147=462$ cm^2.	2+1

34	$\frac{520}{1000} = 0.52$ (1+1). Difference from 0.5 is 0.02 (1) — small because the number of trials is large.	2+1
35	$\angle A + \angle D = 180^\circ$ (co-interior) and $\angle A + \angle C = 180^\circ$ (cyclic) ⇒ $\angle C = \angle D$ (2). Base angles equal ⇒ $AD = BC$ (1).	2+1
	OR: Distance of AB: $\sqrt{17^2 - 15^2} = 8$; of CD: $\sqrt{17^2 - 8^2} = 15$ (2). Same side ⇒ $15 - 8 = 7$ cm (1).	2+1
36	Area at Stage n is $\left(\frac{3}{4}\right)^n$ ⇒ Stage 3: $\frac{27}{64}$ (2). 27 triangles each of area $\frac{1}{64}$ give $\frac{27}{64}$ ✓ (1).	2+1
	OR: $S = 1 + 2 + \cdots + n$ and $S = n + (n-1) + \cdots + 1$ (1). Adding: $2S = n(n+1)$ since there are n pairs, each $n+1$ (1). So $S = \frac{n(n+1)}{2}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) 7 kurtas; left 1 m. (ii) $12.5 \div 2 = 6.25$: 6 kurtas; left 0.5 m $= \frac{1}{2}$ m. (iii) Left: 1 m and $\frac{1}{2}$ m; total $\frac{3}{2}$ m.	2+2+1
	OR: (i) $0.\overline{27} = \frac{27}{99} = \frac{3}{11}$. $x = 1.2\overline{5}$: $10x = 12.\overline{5}$, $100x = 125.\overline{5}$; $90x = 113 \Rightarrow x = \frac{113}{90}$. (ii) $9x = 9 \Rightarrow x = 1$: so $0.\overline{9} = 1$ exactly; decimal forms are not unique.	2+2+1
38	Draw the circle through A, B, D (1). Suppose it cuts CD at E. ABED is cyclic ⇒ $\angle BAD + \angle BED = 180^\circ$ (1). Given $\angle BAD + \angle BCD = 180^\circ$ ⇒ $\angle BED = \angle BCD$ (1). But $\angle BED$ is the exterior angle of $\triangle BEC$ so $\angle BED > \angle BCD$ — a contradiction (2). Hence C lies on the circle.	1+1+1+2
	OR: (i) $A + C = 180$: $8x - 20 = 180 \Rightarrow x = 25$; $B + D = 180$: $6y = 180 \Rightarrow y = 30$ (2). (ii) $\angle A = 80^\circ, \angle B = 70^\circ, \angle C = 100^\circ, \angle D = 110^\circ$ (2). (iii) $80 + 70 + 100 + 110 = 360$ ✓ (1).	2+2+1
39	(i) $\frac{22}{7} \times 56 = 176$ cm (1). (ii) $\frac{176000}{176} = 1000$ (1). (iii) $500 \times 176 = 88000$ cm/min $= 880$ m/min $= 52.8$ km/h (2). (iv) Circumference 220 cm; ratio of distances $= 176 : 220 = 4 : 5$ (1).	1+1+2+1
	OR: (i) $s = 34$; area $= \sqrt{34 \times 17 \times 9 \times 8} = 204$ cm^2 (2). (ii) Altitudes $\frac{2 \times 204}{17} = 24$, $\frac{408}{25} = 16.32$, $\frac{408}{26} \approx 15.69$ cm (2). (iii) The altitude on the shortest side (24 cm) (1).	2+2+1
40	(i) $a + 2d = 12$, $a + 49d = 106 \Rightarrow d = 2$, $a = 8$ (2). (ii) $8 + 28 \times 2 = 64$ (1). (iii) $\frac{50}{2}(8 + 106) = 2850$ (2).	2+1+2
	OR: (i) $S_{10} = 55$ (1). (ii) $S_{25} = 325$ (1). (iii) $\frac{n(n+1)}{2} = 210 \Rightarrow n = 20$ (2). (iv) The 10th triangular number (1).	1+1+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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